SOLAR DESIGN ENGINEERING PACKAGE — DASHBOARD

Current Design	
Project	44755 Savage Rd, Coarsegold CA
Panel	REC Alpha Pure-RX 460
Inverter	Sol-Ark 18K-2P-LV
Battery	Discover Helios ESS
Battery Qty	3
Array Layout	3×12 Portrait x1 Single Mount
DC Array kW	#N/A
Battery Storage kWh	#N/A
Estimated Hardware $	#N/A
Electrical Status	#N/A

Design Status	
Electrical	#N/A
Layout	Column-based strings
Shade	First-pass model
Cost	Hardware estimate only
Mechanical	Engineer review required
Next Step	Validate rack height/wind loading

SOLAR DESIGN ENGINEERING PACKAGE — BUILD PAGE v4.0

Design Inputs		Selected Equipment Values		Live Engineering Summary	
Project	44755 Savage Rd, Coarsegold CA	**Panel W**	#N/A	**Total Panels**	36
Panel Model	REC Alpha Pure-RX 460	**Panel Voc**	#N/A	**DC Array kW**	#N/A
Inverter	Sol-Ark 18K-2P-LV	**Panel Vmp**	#N/A	**String Vmp**	#N/A
Battery	Discover Helios ESS	**Panel Isc**	#N/A	**Cold String Voc**	#N/A
Battery Qty	3	**Panel Imp**	#N/A	**MPPT Operating Current**	#N/A
Orientation	Portrait	**Voc Temp %/C**	#N/A	**MPPT Short-Circuit Current**	#N/A
Mount Type	Single Mount	**Panel Length ft**	#N/A	**Power per MPPT kW**	#N/A
Rows High	3	**Panel Width ft**	#N/A	**Battery Storage kWh**	#N/A
Columns Wide	12	**Panel Cost**	#N/A	**Estimated Hardware $**	#N/A
Banks	1	**Inv MPPT Min**	#N/A	**Cost per DC Watt**	#N/A
Tilt °	37	**Inv MPPT Max**	#N/A	**Panel Width ft**	#N/A
Azimuth °	180	**Inv Max Voc**	#N/A	**Panel Length ft**	#N/A
Design Temp °C	-5	**Inv Max Op A**	#N/A	**Array Width ft**	#N/A
Panels/String	6	**Inv Max Isc A**	#N/A	**Sloped Array Height ft**	#N/A
Total Strings	6	**Inv MPPT Count**	#N/A	**Ground Depth ft**	#N/A
Strings/MPPT	2	**Inv Cost**	#N/A	**Rear Edge Rise ft**	#N/A
MPPT Mode	Manual	**Battery kWh**	#N/A	**Electrical Status**	#N/A
BOS Estimate $	12000	**Battery Cost**	#N/A	**Layout Note**	Row spacing N/A - single structure

Check	Calculated	Limit	Status
String Vmp	#N/A	#N/A	#N/A
Cold Voc	#N/A	#N/A	#N/A
MPPT Current	#N/A	#N/A	#N/A
MPPT Isc	#N/A	#N/A	#N/A
MPPT Count	3.0	#N/A	#N/A

Visual Array Builder — default 3×12 column strings (change string IDs in cells)

S1	S1	S2	S2	S3	S3	S4	S4	S5	S5	S6	S6
S1	S1	S2	S2	S3	S3	S4	S4	S5	S5	S6	S6
S1	S1	S2	S2	S3	S3	S4	S4	S5	S5	S6	S6

String	Physical Zone	MPPT	Shade Pairing Rationale
S1	West edge columns 1-2	MPPT1	Pair west edge with east edge
S2	West-center columns 3-4	MPPT2	Pair symmetric inward zones
S3	Center-west columns 5-6	MPPT3	Pair center strings
S4	Center-east columns 7-8	MPPT3	Pair center strings
S5	East-center columns 9-10	MPPT2	Pair symmetric inward zones
S6	East edge columns 11-12	MPPT1	Pair east edge with west edge

Design rationale: default strings are vertical column groups so east/west tree shadows tend to affect one string group at a time rather than clipping every row simultaneously. MPPT pairs are symmetric west/east to reduce the chance that a single edge shadow drags down both strings on the same tracker.

LAYOUT DESIGNER — Mechanical + Electrical Topology

Inputs linked from Build	
Orientation	Portrait
Rows High	3.0
Columns Wide	12.0
Tilt °	37.0
Mount Type	Single Mount
Array Width ft	#N/A
Sloped Height ft	#N/A
Ground Depth ft	#N/A
Rear Rise ft	#N/A

Panel Map / String Topology

S1	S1	S2	S2	S3	S3	S4	S4	S5	S5	S6	S6
S1	S1	S2	S2	S3	S3	S4	S4	S5	S5	S6	S6
S1	S1	S2	S2	S3	S3	S4	S4	S5	S5	S6	S6

Layout Comparisons

Option	Orientation	Rows	Cols	Panels	Width ft	Rear Rise ft	Ground Depth ft	String Fit	Notes
3×12 single mount	Portrait	3	12	36.0	#N/A	#N/A	#N/A	Clean 6 strings	Compact width; taller rear edge
3×12 single mount	Landscape	3	12	36.0	#N/A	#N/A	#N/A	Clean 6 strings	Lower rear edge; wider footprint
2×18 single mount	Portrait	2	18	36.0	#N/A	#N/A	#N/A	Awkward with 6-panel column strings	Lower but wider; strings may traverse zones
2 banks of 2×9	Portrait	2	9	36.0	#N/A	#N/A	#N/A	Likely awkward string crossing	Needs bank spacing; less desirable electrically

SHADE OPTIMIZER — First Iteration / Conceptual Model

Shade Inputs	
Time Scenario	Morning West Edge
Shadowed Columns	2
Power of shaded panel/string %	15.00%
Unshaded power %	100.00%
String strategy	Column Groups

String Impact Estimate

String	Columns	Shade Hit?	Est Power %	Assigned MPPT
S1	1-2	YES	15%	MPPT1
S2	3-4	NO	100%	MPPT2
S3	5-6	NO	100%	MPPT3
S4	7-8	NO	100%	MPPT3
S5	9-10	NO	100%	MPPT2
S6	11-12	NO	100%	MPPT1

MPPT	String A	String B	Avg Power %	Comment
MPPT1	S1	S6	58%	Symmetric edge pairing
MPPT2	S2	S5	100%	Symmetric mid pairing
MPPT3	S3	S4	100%	Center pairing

This is a first-pass shade model. It is not a production simulator yet. Its purpose is to compare string topologies under edge-shadow behavior. Next iteration can add tree distance/height, solar azimuth, and a time slider.

DESIGN REVIEW CHECKLIST

Area	Check	Status	Notes
Electrical	String Vmp inside MPPT range	#N/A	
Electrical	Cold Voc below inverter max	#N/A	
Electrical	MPPT operating current below max	#N/A	
Electrical	MPPT Isc below max	#N/A	
Layout	Column-based strings selected	PASS	Chosen for east/west edge-shadow tolerance
Layout	3×12 single mount under review	REVIEW	Verify rear-edge height/wind load with engineer
Battery	Battery integration	REVIEW	Confirm Discover/Sol-Ark firmware and wiring kit
Mechanical	Foundation/racking	REVIEW	Requires structural engineer
Site	Tree removal/shading	REVIEW	Measure tree heights and sun window after escrow